Lecture 20

# The non-abelian Gauss law

Maxwell evolution preserves Gauss's law because the divergence of a curl vanishes. In Yang–Mills theory, derivatives become covariant derivatives and no longer commute. We prove that their additional commutators cancel, so the non-abelian constraint also propagates.

**Objectives.** *We derive the temporal-gauge equations with the course normalisation, prove Gauss propagation for smooth fields and compute the constraint algebra. A noncommuting homogeneous example makes the cancellations explicit.*

By the end of this lecture you will be able to:

- Propagate the non-abelian Gauss constraint under smooth evolution.
- Derive the gauge variations and the signed constraint bracket.
- State the scope of the cited Cauchy theorem and residual gauge freedom.

## 20.1 Connection, curvature and pairing

Let $I$ be an open time interval containing zero. We work on $I \times \mathbb{T}^3$, with all coefficients smooth and periodic. Fix $G = \mathrm{SU}(n_c)$, $n_c \geq 2$, and its real Lie algebra $\mathfrak{g} = \mathfrak{su}(n_c)$ of traceless anti-Hermitian matrices.

**Definition 20.1** (Connection and curvature)**.** A *connection* here is the globally defined matrix-valued one-form $A = A_\mu \mathrm{d}x^\mu$, with $A_\mu : I \times \mathbb{T}^3 \to \mathfrak{g}$. Its *curvature* and adjoint covariant derivative are

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \qquad D_\mu X = \partial_\mu X + [A_\mu, X].$$

The *temporal gauge* is $A_0 = 0$.

The bracket is the matrix commutator. The positive real pairing on $\mathfrak{g}$ is $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Choose an orthonormal basis $T_a$, so $\mathrm{tr}(T_a T_b) = -\frac{1}{2}\delta_{ab}$. For SU(2), the Pauli matrices of Lecture 14 give

$$T_a = -\frac{\mathrm{i}}{2}\sigma_a, \qquad [T_a, T_b] = \epsilon_{abc} T_c, \qquad \langle T_a, T_b\rangle = \delta_{ab}.$$

Cyclic invariance of trace proves $\langle [X, Y], Z\rangle = \langle X, [Y, Z]\rangle$. Consequently

$$\partial_i \langle X, Y\rangle = \langle D_i X, Y\rangle + \langle X, D_i Y\rangle. \tag{20.1}$$

On the fundamental cube $Q_{\mathrm{per}} = [0, 2\pi]^3$, the boundary term is $\int_{\partial Q_{\mathrm{per}}} \langle X, Y\rangle n_i \,\mathrm{d}S$. Its opposite faces cancel by periodicity, giving covariant integration by parts. Expansion and the matrix Jacobi identity also give $[D_i, D_j]X = [F_{ij}, X]$.

## 20.2 Temporal Hamiltonian evolution

With coupling $g > 0$, the Yang–Mills action and its scalar density are

$$\begin{aligned} S &= \frac{1}{g^2}\int \operatorname{tr}(F \wedge \star F), \\ L &= \frac{1}{2g^2}\sum_i \langle F_{0i}, F_{0i}\rangle - \frac{1}{4g^2}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle. \end{aligned} \tag{20.2}$$

Spatial indices run from one to three. The double sum counts each spatial curvature pair twice. The action is taken on compact time subintervals. The signature and Hodge star are those of Lecture 9.

Before setting $A_0 = 0$, note that $F_{0i} = \partial_t A_i - D_i A_0$. Thus the momentum and scalar-potential variation are

$$\begin{aligned} \Pi_i &= g^{-2} F_{0i}, \\ \delta_{A_0} S &= -\int \langle \Pi_i, D_i \delta A_0\rangle \operatorname{vol} = \int \langle D_i \Pi_i, \delta A_0\rangle \operatorname{vol}. \end{aligned} \tag{20.3}$$

Here the variation has compact time support, and the spatial face terms from (20.1) cancel. The $A_0$ equation is the Gauss constraint

$$\mathcal{D} := \sum_i D_i \Pi_i = 0. \tag{20.4}$$

It must be derived before removing $A_0$ from the density.

In temporal gauge the spatial Hamiltonian is

$$H = \int_{\mathbb{T}^3} \left[ \frac{g^2}{2}\sum_i \langle \Pi_i, \Pi_i\rangle + \frac{1}{4g^2}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle \right] \mathrm{d}^3x. \tag{20.5}$$

The canonical coordinates are the real components $(A_i^a, \Pi_i^a)$. We use Lecture 7's functional-derivative definition, now on smooth periodic components with their $C^r$ seminorms for every integer $r \geq 0$. Admissible gradients are smooth maps into that same space. The bracket is the sum of the canonical brackets over $i$ and $a$. The constant-matrix closure and Jacobi proof applies unchanged to these finitely many components. All functionals below have such gradients.

For the magnetic part $H_B$ of $H$, use $\delta F_{ij} = D_i \delta A_j - D_j \delta A_i$ and antisymmetry to obtain

$$\delta H_B = \frac{1}{g^2}\int \langle F_{ij}, D_i \delta A_j\rangle \, \mathrm{d}^3x = -\frac{1}{g^2}\int \langle D_i F_{ij}, \delta A_j\rangle \, \mathrm{d}^3x.$$

Thus $H_{A_j} = -g^{-2} D_i F_{ij}$. The periodic face terms cancel as in (20.1). Hamilton's equations are therefore

$$\partial_t A_i = g^2 \Pi_i, \qquad \partial_t \Pi_i = g^{-2}\sum_j D_j F_{ji}. \tag{20.6}$$

These equations are meaningful without imposing (20.4). The propagation theorem explains when they also preserve the missing $A_0$ equation.

## 20.3 Propagation and the constraint bracket

For a fixed smooth periodic parameter $\varepsilon : \mathbb{T}^3 \to \mathfrak{g}$, define the smeared constraint

$$C[\varepsilon] = \int \langle \Pi_i, D_i \varepsilon\rangle \, \mathrm{d}^3x = -\int \langle \mathcal{D}, \varepsilon\rangle \, \mathrm{d}^3x. \tag{20.7}$$

Its parameter is independent of time. The sign agrees with Lecture 11's abelian constraint generator.

**Theorem 20.2** (Non-abelian Gauss law)**.** *Every smooth solution of* (20.6) *on* $I \times \mathbb{T}^3$ *has* $\partial_t \mathcal{D} = 0$*. In particular, zero initial Gauss constraint remains zero. The constraint generator gives*

$$\delta_\varepsilon A_i = D_i \varepsilon, \qquad \delta_\varepsilon \Pi_i = [\Pi_i, \varepsilon]. \tag{20.8}$$

*For fixed smooth periodic* $\varepsilon$ *and* $\zeta$*, the brackets satisfy*

$$\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]], \qquad \{C[\varepsilon], H\} = 0. \tag{20.9}$$

*Thus the Gauss constraints are first class.*

*Proof.* Differentiate the constraint, including the time dependence of $D_i$:

$$\partial_t \mathcal{D} = \sum_i D_i \partial_t \Pi_i + \sum_i [\partial_t A_i, \Pi_i] = g^{-2} \sum_{i,j} D_i D_j F_{ji} + g^2 \sum_i [\Pi_i, \Pi_i].$$

The last commutators vanish individually. Antisymmetry of $F$ and interchange of the two summed indices give

$$2 \sum_{i,j} D_i D_j F_{ji} = \sum_{i,j} [D_i, D_j] F_{ji} = \sum_{i,j} [F_{ij}, F_{ji}] = 0.$$

This proves propagation without assuming the constraint during the calculation.

Next, vary $C[\varepsilon]$ at a fixed time. Invariance of the pairing gives

$$\delta C[\varepsilon] = \int \left[ \langle \delta \Pi_i, D_i \varepsilon \rangle + \langle \Pi_i, [\delta A_i, \varepsilon] \rangle \right] \mathrm{d}^3 x,$$
$$C_{\Pi_i} = D_i \varepsilon, \qquad C_{A_i} = [\varepsilon, \Pi_i].$$

Hamiltonian vector fields therefore give (20.8). These are statements about the gradients and smeared observables, rather than brackets of inadmissible point-evaluation functionals.

Substitute these gradients into the canonical bracket. Pairing invariance and the derivation rule for $D_i$ give

$$\begin{aligned} \{C[\varepsilon], C[\zeta]\} &= \int \left[ \langle [\varepsilon, \Pi_i], D_i \zeta \rangle - \langle D_i \varepsilon, [\zeta, \Pi_i] \rangle \right] \mathrm{d}^3 x \\ &= - \int \langle \Pi_i, [\varepsilon, D_i \zeta] + [D_i \varepsilon, \zeta] \rangle \, \mathrm{d}^3 x = -C[[\varepsilon, \zeta]]. \end{aligned}$$

Finally, the propagation calculation is an identity after substitution of the Hamiltonian vector field at any phase point. It needs no existence claim for a path through that point. Pairing this zero directional derivative of $\mathcal{D}$ with the fixed parameter gives $\{C[\varepsilon], H\} = 0$ strongly. The first bracket generally vanishes only on the constraint surface. This is the first-class condition of Lecture 11, rather than strong commutativity of non-abelian constraints. ∎

## 20.4 Noncommuting homogeneous data

Take SU(2), $g = 1$ and spatially constant data on the torus:

$$A_1 = T_1, \quad A_2 = T_1 + T_2, \quad A_3 = 0, \qquad \Pi_1 = T_1 + T_2, \quad \Pi_2 = T_1, \quad \Pi_3 = 0.$$

The curvature is nonzero: $F_{12} = T_3$. Both momenta displayed are nonzero. The two nonzero Gauss terms cancel: $[A_1, \Pi_1] = T_3$ and $[A_2, \Pi_2] = -T_3$. For spatially constant fields the evolution reduces to

$$\partial_t A_i = \Pi_i, \qquad \partial_t \Pi_i = \sum_j [A_j, [A_j, A_i]].$$

At the stated instant, the two nonzero derivatives are

$$\partial_t \Pi_1 = [T_1 + T_2, -T_3] = -T_1 + T_2,$$
$$\partial_t \Pi_2 = [T_1, T_3] = -T_2, \qquad \partial_t \Pi_3 = 0.$$

The kinetic contributions $[\partial_t A_i, \Pi_i]$ vanish. The other contributions are genuinely nonzero: $[A_1, \partial_t \Pi_1] = T_3$ and $[A_2, \partial_t \Pi_2] = -T_3$. Thus the Gauss derivative cancels as claimed. The Hamiltonian density is $3/2 + 1/2 = 2$, so this is not a zero-field example.

## 20.5 Cited Cauchy theory

The theorem above is conditional on smooth evolution. It does not construct an energy-class Yang–Mills solution. For comparison, the following local result on $\mathbb{R}^3$, rather than the torus, is *cited*. Tao, *Local well-posedness of the Yang–Mills equation in the temporal gauge below the energy norm*, Theorem 1.1, p. 2, arXiv:math/0005064, proves: for $s > 3/4$, sufficiently small data $(A(0), \partial_t A(0)) \in H^s \times H^{s-1}$ satisfying $D_i \partial_t A_i(0) = 0$ have a solution on $[-1, 1]$. It lies in a Banach space $X$ constructed in that paper, with $X \subset C_t H^s \cap C_t^1 H^{s-1}$, is unique in $X$, and depends continuously on the data into $X$. Here $\|u\|_{H^r} = \|\langle k \rangle^r \widehat{u}\|_2$, summed over components, with negative exponents interpreted distributionally. Tao uses the opposite metric signature; reversing the whole metric leaves $D_\mu F^{\mu\nu} = 0$ unchanged because both curvature indices are raised.

Smooth-data existence on $\mathbb{R}^{1+3}$ is treated by Eardley–Moncrief (1982), *Communications in Mathematical Physics* 83, 171–191, Part I. Finite-energy theory on $\mathbb{R}^{1+3}$ is treated by Klainerman–Machedon (1995), *Annals of Mathematics* 142, 39–119, Finite energy solutions. These analytic results are cited; none is inferred from constraint propagation.

## 20.6 Residual gauge and global gauge choices

The finite gauge transformation has the convention $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$, with adjoint fields transforming by $\Pi_i' = h^{-1}\Pi_i h$. In temporal gauge $A_0' = h^{-1}\partial_t h$. It preserves $A_0 = 0$ exactly when $\partial_t h = 0$. The generators above therefore describe time-independent residual gauge transformations.

The Coulomb condition $\partial_i A_i = 0$ also does not select a unique representative globally. On the torus, take $h(x) = \exp(2x^1 T_3)$ in SU(2). It is periodic since $\exp(4\pi T_3) = I$. It sends $A = 0$ to $A_1' = 2T_3$, $A_2' = A_3' = 0$. Both potentials satisfy the Coulomb condition and have zero curvature. The periodic object here is group-valued $h$, rather than a single-valued real gauge parameter of Lecture 12. This is a *Gribov copy*: distinct gauge-equivalent connections obeying the same gauge condition.

A stronger global obstruction is *cited*: Singer, *Some remarks on the Gribov ambiguity*, *Communications in Mathematical Physics* 60 (1978), 7–12, doi:10.1007/BF01609471, treats principal bundles over the four-sphere with compact non-abelian structure group and rules out a continuous choice of one connection on each orbit in that setting. This global obstruction does not rule out local gauge slices. Lecture 21 studies the coupling to a scalar field near a vacuum.

**Summary.** For smooth Yang–Mills evolution on the torus in temporal gauge, $\Pi_i = g^{-2}F_{0i}$ and the constraint $D_i\Pi_i = 0$ propagates. Antisymmetry of the curvature cancels the commutator of covariant derivatives. The smeared generator gives $\delta A_i = D_i\varepsilon$ and $\delta\Pi_i = [\Pi_i, \varepsilon]$, with $\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]]$. Existence at low regularity requires the separately cited analytic theorems. Temporal gauge retains time-independent gauge freedom, and global Coulomb uniqueness cannot be assumed.

**Self-check.** Why does differentiating $D_i\Pi_i$ give a term in addition to $D_i\partial_t\Pi_i$? What happens to it in temporal Hamiltonian evolution?

**Answer.** The connection in $D_i$ also varies, giving $[\partial_t A_i, \Pi_i]$. Since $\partial_t A_i = g^2\Pi_i$, each such term is $g^2[\Pi_i, \Pi_i] = 0$.

**Self-check.** For the homogeneous data, what remains of $\partial_t\mathcal{D}$ if the contribution from $i = 2$ is mistakenly omitted?

**Answer.** The uncancelled contribution is $T_3$. Both original Gauss terms and both magnetic derivative terms must be retained.

**Self-check.** Does first class mean that $\{C[\varepsilon], C[\zeta]\}$ vanishes identically? How does the abelian limit compare with Lecture 11?

**Answer.** The bracket is $-C[[\varepsilon, \zeta]]$, so it vanishes on the constraint surface but need not vanish off it. For an abelian algebra the parameter commutator is zero, giving the strongly commuting constraints of Lecture 11.

**Self-check.** Does Tao's stated theorem give uniqueness among every solution in $C_t H^s \cap C_t^1 H^{s-1}$ for arbitrary large data? Which gauges preserve $A_0 = 0$ after it has been imposed?

**Answer.** The stated result assumes sufficiently small compatible data and gives uniqueness in its specified space $X$. Residual finite transformations satisfy $\partial_t h = 0$.